

Group Assignment: GenAI-Assisted Career and Operating Systems Exploration for Cooperative Education / Job Training

1. Assignment Purpose

Junior students are approaching an important stage of their study in which they will soon participate in a Cooperative Education (COOP) program and/or job training at an organization of their interest, maybe outside the university.

Before applying, students should begin considering:

- What job position would I like to experience?
- What type of organization would I like to work with?
- What does that position actually involve?
- What knowledge and skills are expected for that position?
- What operating system(s), hardware, infrastructure, software, and tools are commonly associated with that work?
- What should I learn and practice before applying?
- How can Generative AI (GenAI) assist me in preparing for and performing real-world work?

Since this assignment is part of an Operating Systems course, particular attention must be given to the relationship among the selected job position, **operating systems**, **computer hardware**, **system architecture/infrastructure**, and **software tools/applications**.

The assignment is intended to help students connect Operating Systems concepts learned in class with realistic computing environments that they may encounter during COOP, job training, or future employment.

At the same time, students will gain practical experience in using different **Generative AI platforms/models** for knowledge exploration and professional work.

2. Group Formation and Submission

2.1 Form a group of 3–5 members.

2.2 Group members may come from different sections of the course.

2.3 Each group should collaboratively select:

- one **target job title/position**, and
- one **company/organization** of interest.

2.4 The selected position should involve computing systems sufficiently for the group to investigate its relationship with **Operating Systems**, **hardware/infrastructure**, and **software/tools**. Examples of possible positions are listed in **Appendix A**.

Each group must submit a 10–15 page (excluding the cover page) report in PDF format, using 12-point Times New Roman font on an A4 layout. Only the *group representative* must submit a *single copy* of the report to **204341 Final Assignment Submission in the CMU Mango System** by **Sunday, October 18, 2026**, at **23:59**.

The following information for each group member must be clearly written on the *cover page*:

- 1) Student ID
- 2) Full name
- 3) Mobile phone number (Kindly requested solely for contact regarding emergencies or grading issues)
- 4) Section (001 or 002)



The selected organization should preferably be one in which students could realistically consider undertaking COOP, job training, an internship, or future employment.

3. Required Use of Generative AI

Each group must select and use **at least TWO** GenAI platforms in completing this assignment. Examples of possible platforms are listed in **Appendix B**.

Students should consider the strengths of different GenAI platforms when deciding which tool to use. For example, a general-purpose GenAI platform may assist with knowledge exploration, technical explanations, troubleshooting, command examples, configuration examples, programming, or comparing technologies, whereas a source-grounded platform such as NotebookLM may assist with synthesizing information from a collection of documents supplied by the students and generating summaries or presentation materials from those sources.

4. Task 1: Knowledge Exploration

Use appropriate GenAI platforms/models to assist your group in exploring the following areas.

4.1 Target Job Position

The investigation on the selected position should focus particularly on what a Computer Science student preparing for COOP/job training should understand and practice before applying. Examples of information to be investigated for the selected position are listed in **Appendix C**.

4.2 Hard Skills and Technical Knowledge

Identify the important technical or hard skills required or recommended for the selected position. Examples of hard skills are listed in **Appendix D**.

4.3 Operating Systems, Hardware, and Infrastructure Exploration

Students must investigate the computing environment associated with their selected job position and organization. Not only is the purpose to state that “Windows” or “Linux” is used, but students should also investigate, for example, why a particular OS is appropriate, what hardware/infrastructure it operates on, how it is configured, and what software or tools operate within that environment.

Examples of the system architecture/infrastructure that students may encounter, regarding to the selected position, are available in **Appendix E**.

4.4 Software, Tools, and Applications Exploration

Investigate the important software, tools, and applications that operate on or interact with the identified operating system(s).

Examples may include:

- Development tools
E.g., VS Code, Android Studio, Xcode, JupyterLab, IDEs and compilers, etc.
- Programming and runtime environments
E.g., Python, C/C++, .NET, CUDA, JavaScript/Node.js, etc.
- System and DevOps (software development and IT operations) Tools
E.g., Git, Docker, Kubernetes, SSH, Shell scripting, Bash, CI/CD tools, system monitoring tools, etc.
- Server and Database Software
E.g., Apache, MySQL, PostgreSQL, MongoDB, Redis, etc.

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- AI/Data Tools
E.g., PyTorch, TensorFlow, CUDA, Jupyter, data-processing frameworks, etc.

4.5 Connecting the Job to Operating Systems Knowledge

Based on the exploration, identify Operating Systems concepts from the course that are relevant to the selected job.

For example:

AI/ML Engineer → Linux → GPU workstation → CUDA/PyTorch → process/GPU memory/resource management
DevOps Engineer → Linux → cloud servers → Docker/Kubernetes → processes, networking, virtualization, containers, security
Mobile Developer → Android → ARM-based smartphone → Android Runtime/SDK → processes, memory, permissions, file systems, device management

4.6 Soft Skills

Where applicable, investigate important soft skills. Then, explain why the identified skills are relevant to the selected position and working environment. Examples of important soft skills are available in [Appendix F](#).

4.7 Organization Exploration

Investigate the selected organization sufficiently to understand the environment in which the selected position may operate. Examples of relevant information are available in [Appendix G](#).

Do not claim that an organization uses a particular OS, hardware platform, or technology unless reliable evidence is available. If the actual internal infrastructure cannot be determined, students may instead investigate a **reasonable or typical computing environment for that type of position**, clearly identifying it as a recommendation or assumption rather than a verified fact about the organization.

5. Task 2: GenAI-Assisted Report Creation

After completing the knowledge exploration, prepare a professional report presenting and explaining the findings.

At least ONE GenAI platform must be used to assist with the report-creation process.

Examples of work GenAI may assist with are listed in [Appendix H](#).

At least one visualization should explicitly demonstrate the relationship among:

Job Position → Organization/Workload → Operating System → Hardware/Infrastructure → Software/Tools → Required Knowledge/Skills
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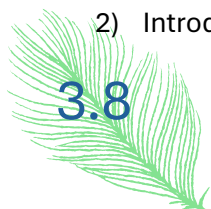
Students remain responsible for verifying, correcting, and organizing all GenAI-assisted content.

6. Suggested Report Structure

The report should contain the following topics*, but not limited to:

*Note that any topic that ultimately could **not** be found must be clearly noted.

- 1) Cover Page
- 2) Introduction and Career Interest



- 3) Selected Job Position and Job Description
- 4) Hard Skills and Soft Skills
- 5) Selected Organization and Working Environment
- 6) Operating System(s) Required or Recommended
- 7) Hardware Requirements
- 8) System Architecture/Infrastructure
- 9) Software, Tools, and Applications
- 10) Relationship to Operating Systems Course Knowledge
- 11) Use of Generative AI Platforms/Models
- 12) Conclusion
- 13) References
- 14) Appendix: Complete GenAI Prompts/Input Messages

7. References – REQUIRED

All information sources used for the assignment must be explicitly acknowledged in the References section. The References section must appear **before** the Appendix. Examples of possible references that may be included are listed in [Appendix I](#).

For every online reference, clearly provide the **date** on which the source was accessed/searched. For GenAI, identify the platform whenever available.

Example:

OpenAI. ChatGPT, [specific model name/version]. Used for career, OS, hardware, and software exploration. Accessed: 15 October 2026.

8. Appendix – Complete GenAI Prompts / Input Messages – REQUIRED

The Appendix must contain the complete prompts/input messages submitted to the GenAI platforms/models for both:

- Knowledge Exploration, and
- Report Creation.

For each significant GenAI interaction, identify:

- GenAI platform [with model/version]
- Purpose of interaction
- Complete prompts/input messages, **essential/significant** to the purpose
- Supporting documents/data supplied to the model, if applicable

Screenshots are allowed as supporting evidence, provided that they are *clear and readable*.

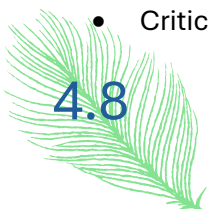
However, all **significant** prompts/input messages used to produce the submitted work **must be completely shown**. Do not provide only shortened, rewritten, or summarized prompts.

Failure to provide complete prompts/input messages may result in a substantial deduction of the assignment score.

9. Responsible Use of GenAI

Students must:

- Critically examine GenAI-generated information.



- Verify important technical and organizational facts.
- Distinguish verified facts from GenAI recommendations or assumptions.
- Compare information from multiple sources where appropriate.
- Correct inaccurate, incomplete, misleading, or outdated information.
- Clearly acknowledge all GenAI platforms/models used.
- Keep complete records of significant prompts.
- Avoid submitting confidential information, passwords, private data, proprietary source code, or other sensitive organizational information to public GenAI services.

Simply copying and pasting GenAI-generated text **without** understanding, verification, or appropriate modification **does not satisfy the objectives of this assignment**.

10. Expected Learning Outcomes

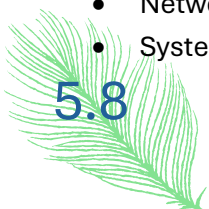
After completing the assignment, students should be able to:

- 1) Identify a potential career position for future COOP/job training.
- 2) Explain the responsibilities and expectations associated with that position.
- 3) Identify relevant hard skills and soft skills.
- 4) Identify the operating system(s) commonly associated with the selected type of work.
- 5) Explain relationships among OS, hardware architecture, infrastructure, applications, and job requirements.
- 6) Identify software/tools that should be learned for the selected job.
- 7) Connect Operating Systems concepts studied in class with a realistic workplace environment.
- 8) Develop a practical hands-on preparation plan involving OS installation, configuration, software/tools, and relevant hardware.
- 9) Use multiple GenAI platforms/models appropriately for different tasks.
- 10) Critically evaluate and verify GenAI-generated technical information.
- 11) Properly document GenAI use, prompts, and external sources.
- 12) Develop greater awareness of how Operating Systems knowledge and GenAI skills can contribute to professional work in a real computing environment.

Appendix

A. Examples of possible *positions* include:

- Software Developer / Software Engineer
- Backend Developer
- Web Developer
- Mobile Application Developer
- Data Analyst
- Data Scientist
- AI / Machine Learning Engineer
- Computer Vision Engineer
- Cybersecurity Analyst
- Network Engineer
- System Administrator



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- Cloud Engineer
- DevOps Engineer
- Site Reliability Engineer
- Database Administrator
- Game Developer
- Embedded Systems Developer
- IoT Developer
- IT Support Specialist
- Other appropriate Computer Science or IT positions

B. Examples of possible *GenAI platforms* include, but are not limited to:

Note that the **free** versions of the platforms are sufficient for this assignment's purpose.

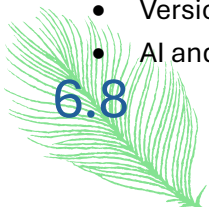
- ChatGPT
- Google Gemini
- NotebookLM
- Claude
- Microsoft Copilot
- Perplexity
- GitHub Copilot

C. Examples of the selected *position's information* include:

- Job title
- Typical job description
- Major duties and responsibilities
- Typical daily or weekly tasks
- Types of projects that may be assigned
- Educational background normally expected
- Previous experience, portfolio, certificates, or other qualifications that may be advantageous
- Hardware, operating systems, software, and technical tools commonly associated with the position

D. Examples of *hard skills* include:

- Programming languages
- Operating systems
- Computer architecture
- Computer networks
- Database systems
- Cloud computing
- Virtualization
- Containers
- Cybersecurity
- Software development tools
- Version control
- AI and machine learning



- Data analysis
- Web/mobile technologies
- DevOps
- Computer graphics
- Image processing/computer vision
- Embedded systems
- Other position-specific technologies

E. Examples of the *system architecture/infrastructure* that students may encounter include:

- Client-server architecture
- Web servers
- Application servers
- Database servers
- File servers
- Cloud infrastructure
- Virtual machines
- Hypervisors
- Containers
- Container orchestration
- Distributed systems
- Clustered systems
- GPU servers
- High-performance computing
- Edge computing
- Embedded systems
- IoT infrastructure
- Network-attached storage
- Backup systems

F. Examples of important *soft skills* include:

- Communication
- Teamwork
- Problem solving
- Critical thinking
- Time management
- Presentation
- Technical writing
- Adaptability
- Professional responsibility
- English or other language skills

G. Examples of the selected *organization's relevant information* include:

- Organization background



- Main products/services
- Customers or target markets
- Technologies associated with the organization
- IT/software/data/AI activities
- Computing infrastructure, where publicly available
- Working environment or organizational culture, where reliable information is available
- Projects/products related to the selected position
- Existing or previous internship, COOP, or job opportunities

H. Examples of *report work* that GenAI may assist with include:

- Organizing information
- Summarizing findings
- Improving explanations
- Creating tables
- Creating conceptual illustrations
- Creating mind maps
- Creating charts
- Creating system architecture diagrams
- Creating OS/hardware/software relationship diagrams
- Creating career-preparation roadmaps
- Improving language and readability

I. Examples of *references* include:

- All GenAI platforms and models used
- Organization websites
- Job advertisements/job descriptions
- OS documentation
- Hardware documentation
- Software/tool documentation
- Technical websites
- Academic publications
- Online articles/videos
- Other information sources

